

Spatiotemporal Noise Stabilizes Diversity in Strongly-Competitive Metacommunities

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Joint work with **Daniel Swartz**

In collaboration with [Jeff Gore](#) & [Mehran Kardar](#)

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Natural ecosystems are diverse

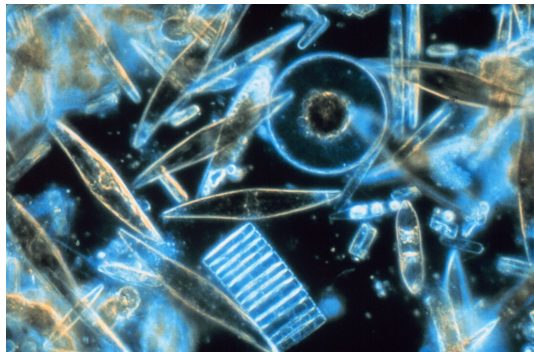
Coral reef: $\mathcal{O}(10^4)$ coexisting species

(Knowlton et al., 2010)

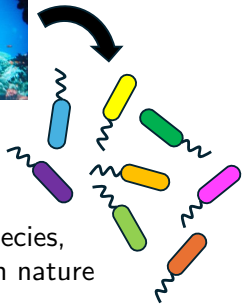


“Paradox of Plankton”: $\mathcal{O}(10^3)$ species

(Hutchinson, 1961)



Artificial ecosystems are not as diverse

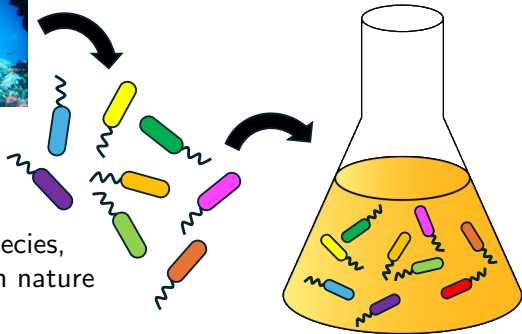


$S \gg 1$ species,
coexisting in nature

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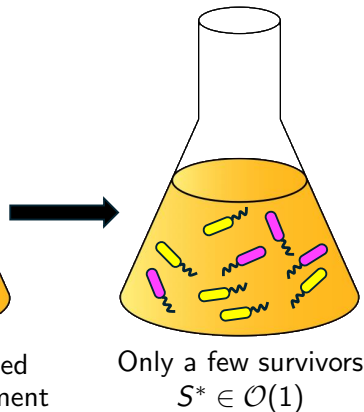
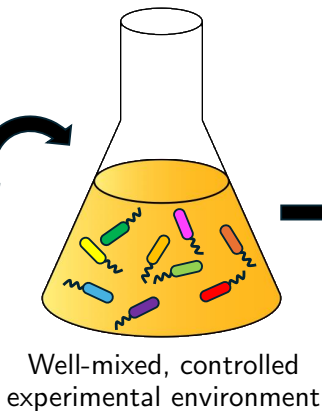
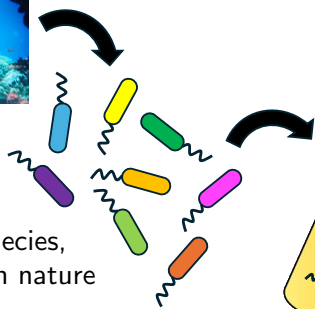


Well-mixed, controlled
experimental environment

Artificial ecosystems are not as diverse



$S \gg 1$ species,
coexisting in nature



(Goldford et al, 2018; Dal Bello et al, 2020; Hu et al., 2022)

Theoretical ecosystems are no better

- ▶ Abundances $\{N_1, \dots, N_S\}$ evolve as

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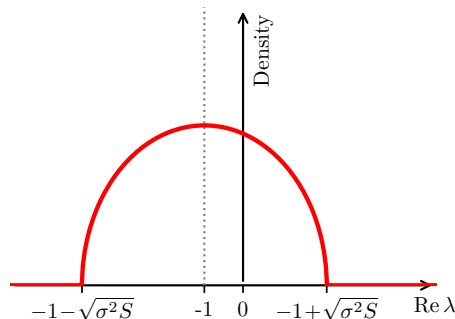
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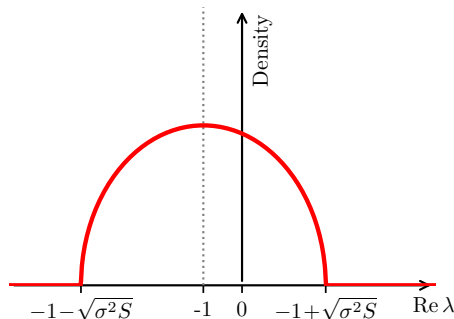
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 $\Rightarrow$ **“Diversity-stability paradox”**

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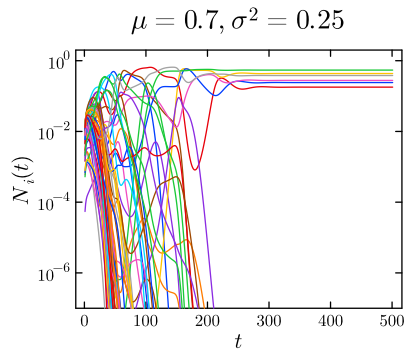
- **Generalized Lotka-Volterra model (GLV):**

$$\dot{N}_i = rN_i \left[1 - N_i - \sum_{j \neq i}^S A_{ij} N_j \right], \quad A_{ij} \sim \text{i.i.d. } \mathcal{N}(\mu, \sigma^2)$$

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Diverse initial condition



Low diversity fixed point

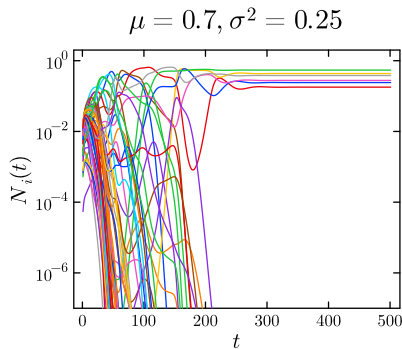
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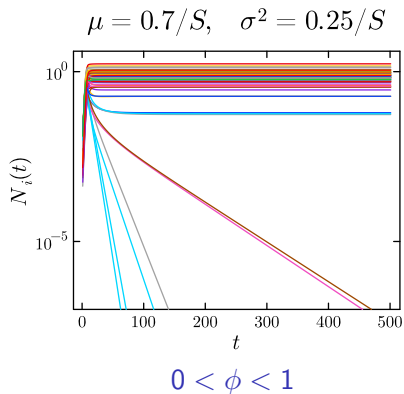
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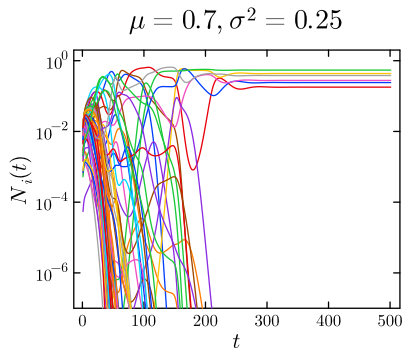
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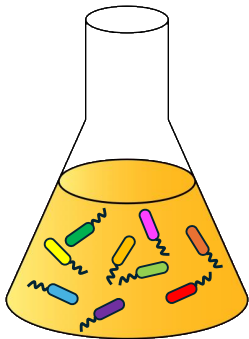
But real communities are strongly-interacting $\mu, \sigma^2 \in \mathcal{O}(1)$

(Hu et al, 2022; Dal Bello et al, 2020; Hutchinson, 1961).



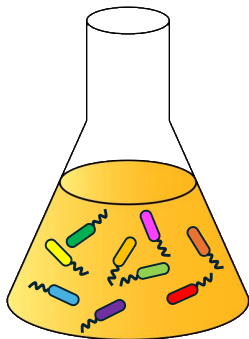
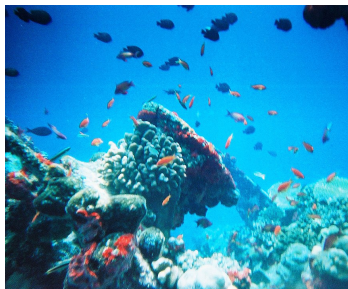
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Real environments are:

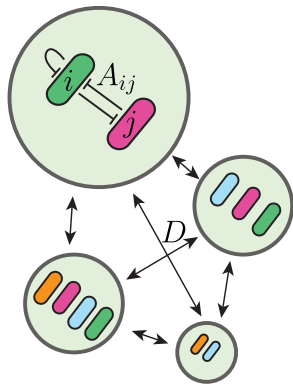
- ▶ **Spatially extended** (Roy et al.; Lorenzana et al.; Pearce et al.; many others)
- ▶ **Fluctuating** (Chesson, 1994; Descamps-Julien & Gonzalez, 2005; many others)

Outline

1. Model: GLV metacommunity with spatiotemporal noise
2. Noise-induced transition to coexistence
3. Low-noise phase: effective GLV dynamics
4. High-noise phase
 - Power law abundance distribution
 - Mapping to θ -GLV
 - Cavity solution of θ -GLV model

Implementing space: Ecological metacommunities

- ▶ Network of P **patches**: $\alpha = 1, \dots, P$
- ▶ Migration at rate D to neighbors $\partial\alpha$
- ▶ **Local** competition within each patch

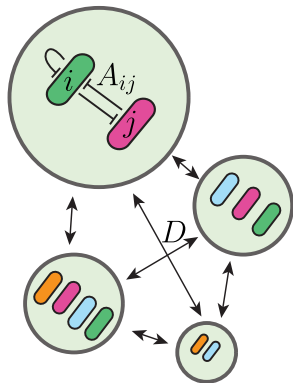


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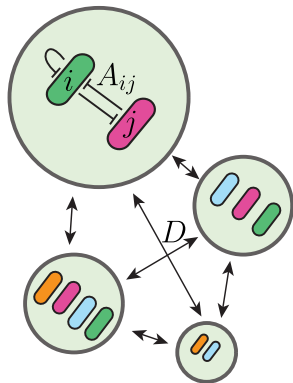
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This talk: *fully connected, infinite networks* $P \rightarrow \infty$.



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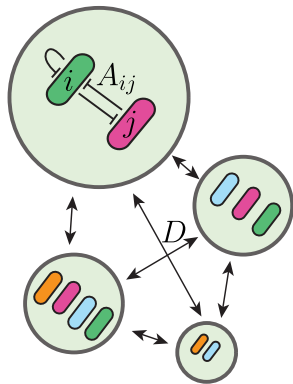
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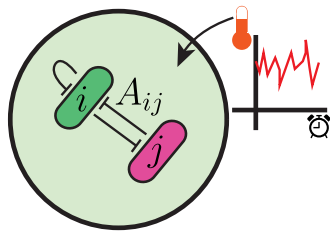
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$$\langle N_i \rangle \equiv \frac{1}{P} \sum_{\alpha} N_{i\alpha}$$



Implementing fluctuations: environmental noise

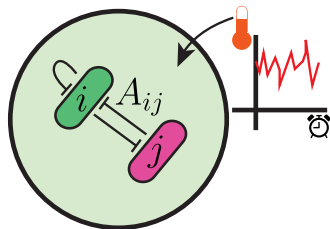
$$\dot{N}_i = rN_i \left[1 - N_i - \sum_{j \neq i} A_{ij} N_j \right]$$



Implementing fluctuations: environmental noise

$$\dot{N}_i = rN_i \left[g(t) - N_i - \sum_j A_{ij} N_j \right]$$

- Bare **growth rate** fluctuates in time

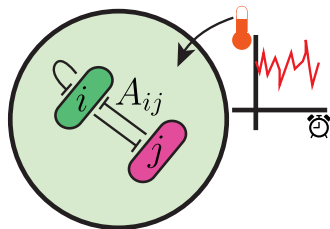


Implementing fluctuations: environmental noise

$$\dot{N}_i = rN_i \left[1 - N_i - \sum_j A_{ij} N_j \right] + \sqrt{2T} N_i \eta_i(t)$$

- Bare growth rate fluctuates in time
- **Minimal choice:** white noise of strength T

$$\langle \eta_i(t) \eta_j(t') \rangle = \delta_{ij} \delta(t - t')$$



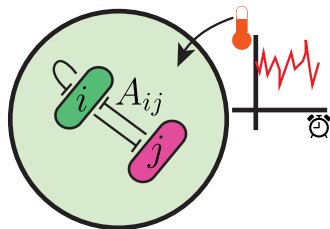
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- ▶ **To model extinction:**
Cutoff $N_c \ll 1$ below which $N(t)$ is set to zero



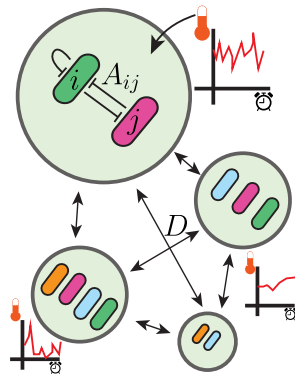
Metacommunity with spatiotemporal noise

- **Space:** $P \gg 1$ patches, migration at rate D .
- **Competition:** local GLV dynamics
- **Fluctuations:** white noise in growth rates

$$\dot{N}_{i\alpha} = rN_{i\alpha} \left[1 - N_{i\alpha} - \sum_{j \neq i} A_{ij} N_{j\alpha} \right] + D[\langle N_i \rangle - N_{i\alpha}] + \sqrt{2T} N_{i\alpha} \eta_{i\alpha}(t),$$

Species: $i = 1, \dots, S$, $\langle \eta_{i\alpha}(t) \eta_{j\beta}(t') \rangle = \delta_{ij} \delta_{\alpha\beta} \delta(t - t')$

Patches: $\alpha = 1, \dots, P$,



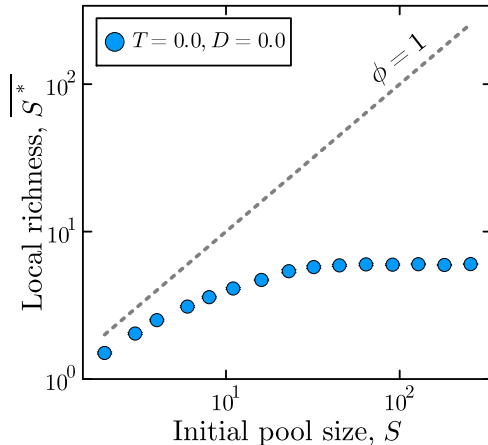
Strongly-interacting metacommunity with environmental noise

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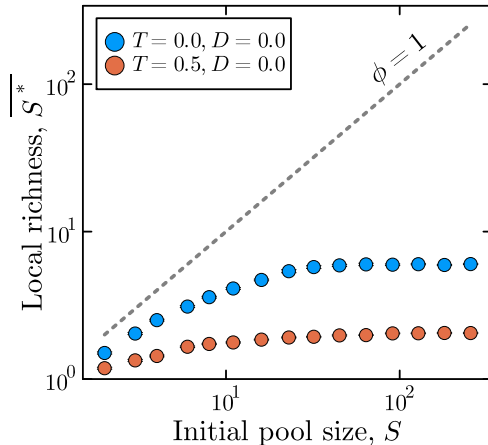
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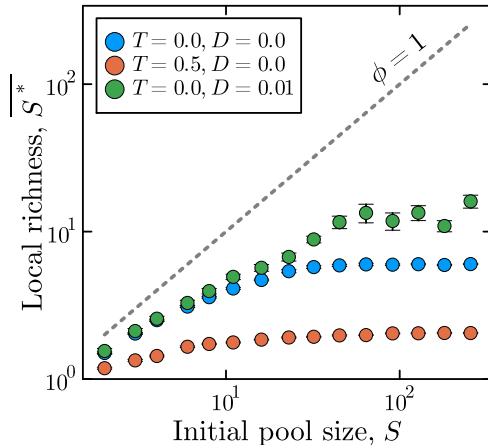
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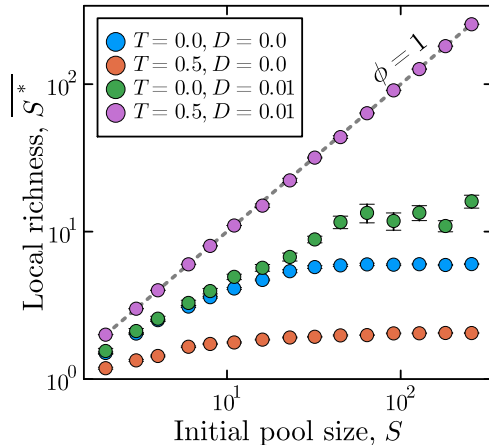
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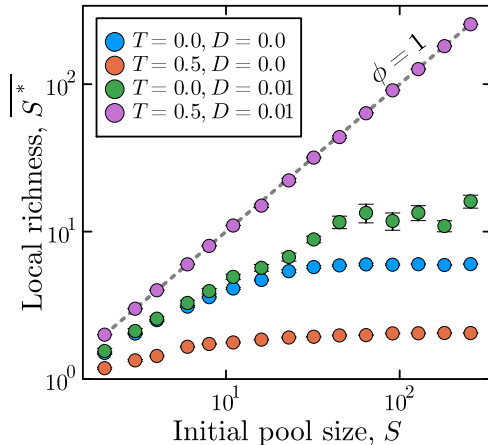


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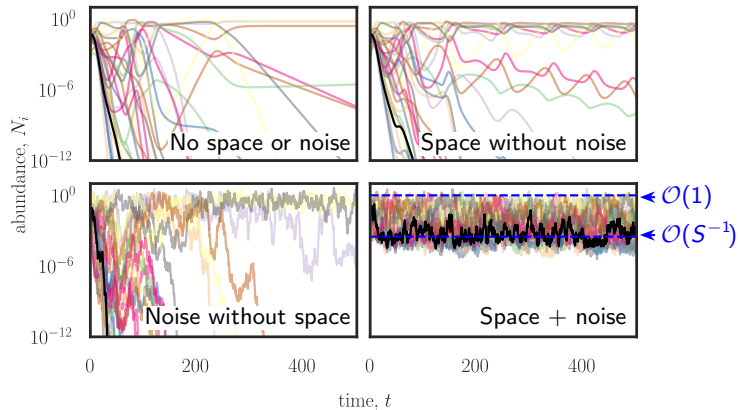
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**Spatiotemporal noise stabilizes
arbitrarily rich metacommunities**



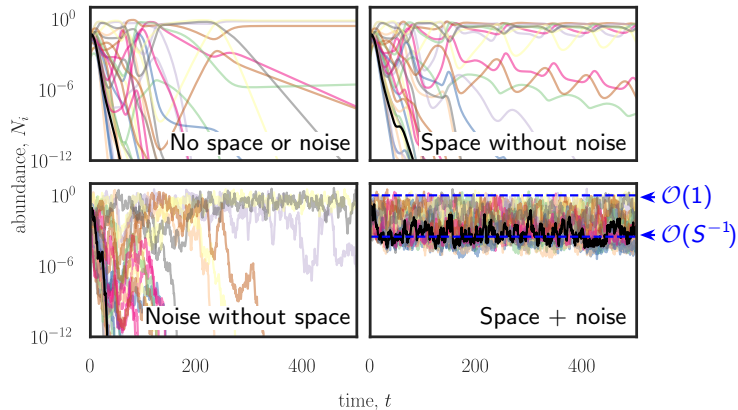
Spatiotemporal noise stabilizes arbitrarily rich metacommunities

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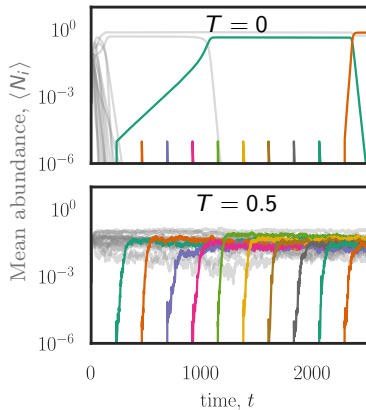


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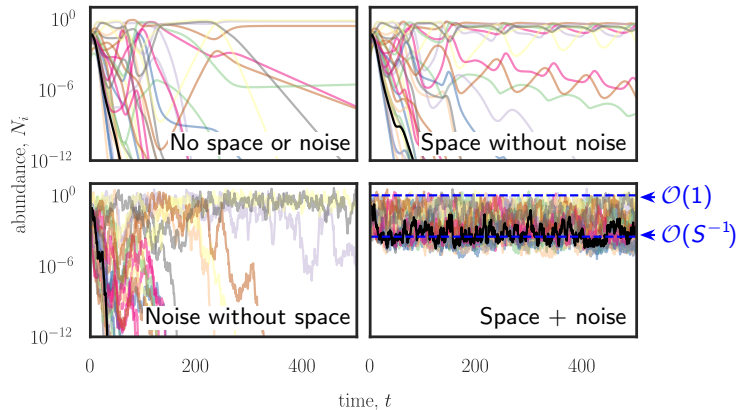


Sequential invasion:

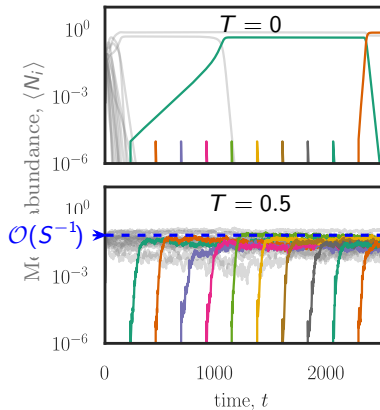


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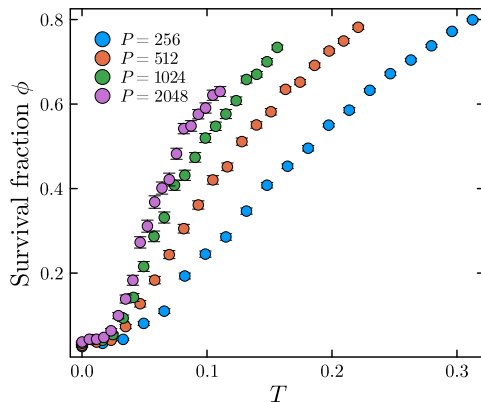


Note: Mean abundances are $\mathcal{O}(S^{-1})$.

Noise-induced transition from $\phi = 0$ to $\phi = 1$

- ▶ Critical temperature T_c separates **competitive exclusion** and **coexistence**.
- ▶ Transition sharpens as $P \rightarrow \infty$:

$$\lim_{P \rightarrow \infty} \phi(T) = \begin{cases} 1, & (T > T_c) \\ 0, & (T < T_c) \end{cases}$$



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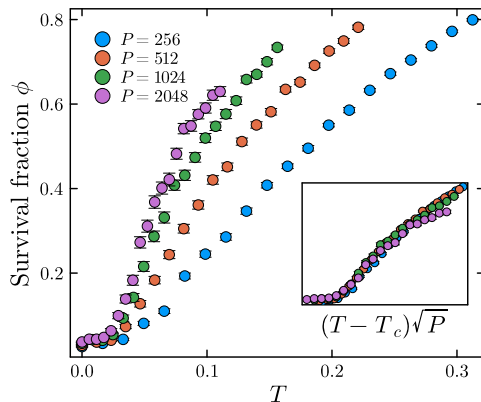
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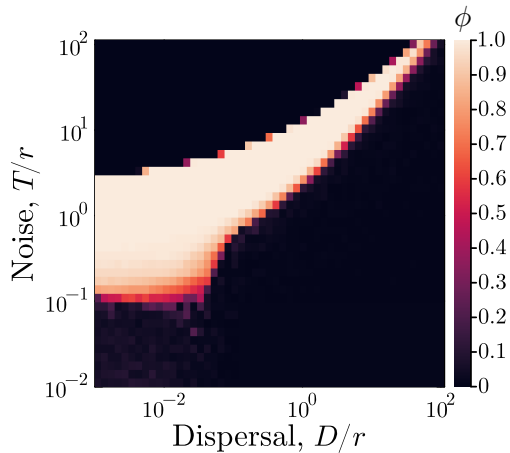
- ▶ Finite-size scaling form

$$\phi(T) = f \left[(T - T_c) \sqrt{P} \right]$$

where $f(-\infty) = 0, f(\infty) = 1$.

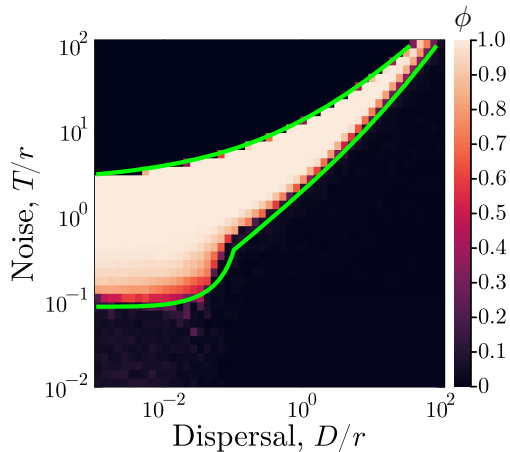


Phase diagram in the D, T plane



Phase diagram in the D, T plane

We identify 3 phases:

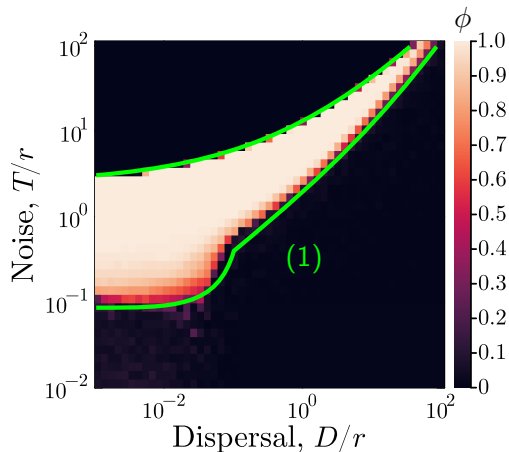


Phase diagram in the D, T plane

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(1) **Competitive exclusion:** $S^* \in \mathcal{O}(1)$, $\phi \simeq 0$.

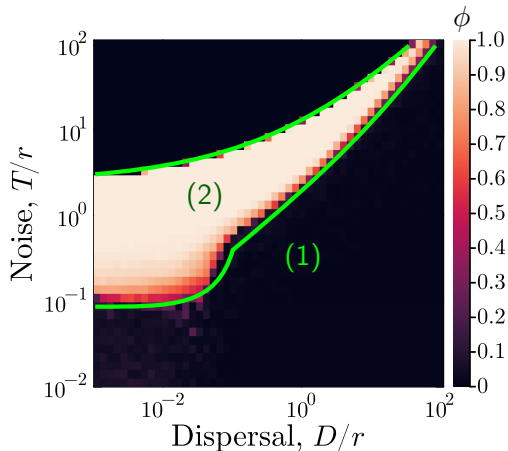
(Similar to single-patch GLV)



Phase diagram in the D, T plane

We identify 3 phases:

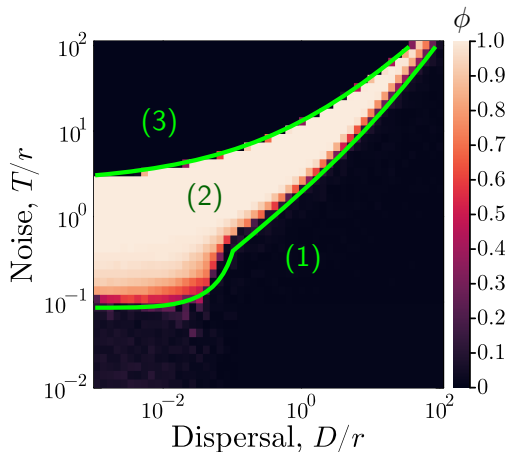
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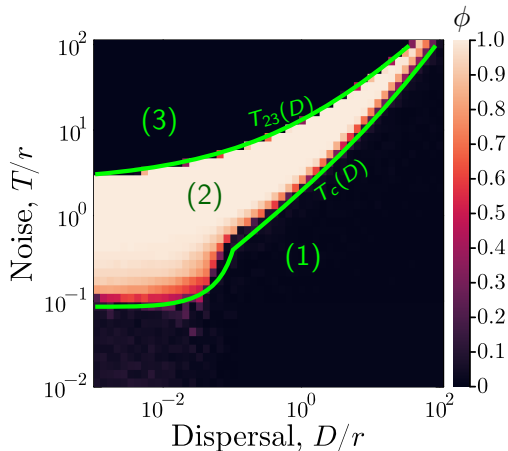
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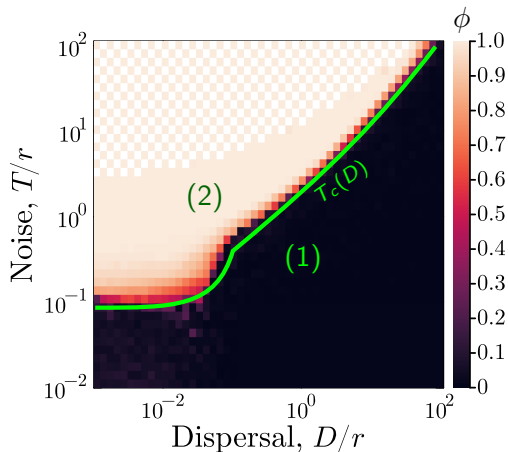
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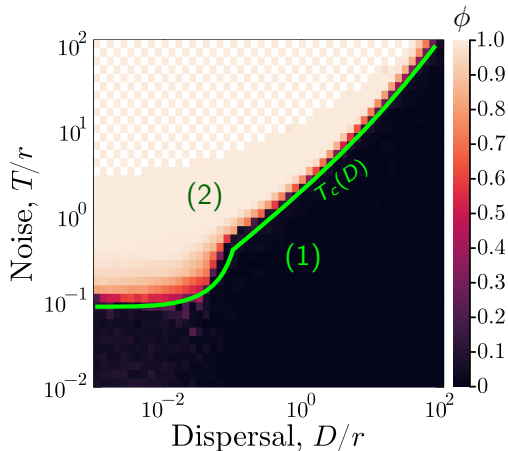
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- 2. $T_c(D) \propto D$ for $D \gg r$



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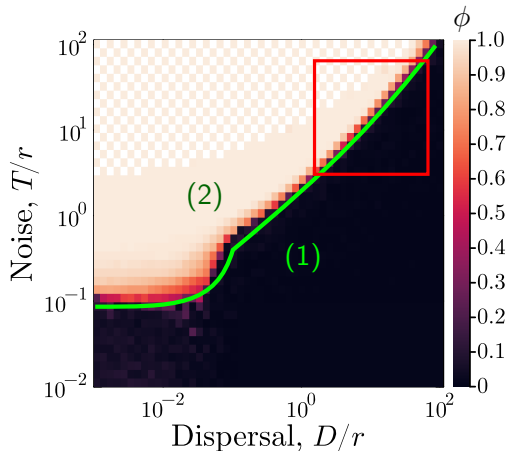
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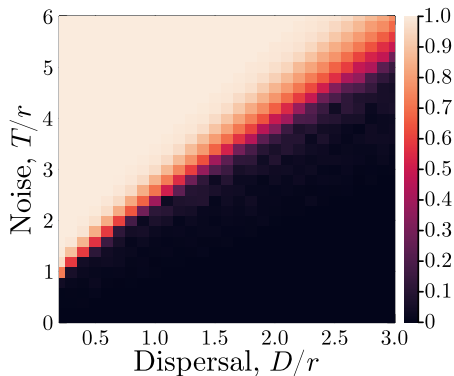
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$D \gg r \Leftrightarrow$ many migration events per generation time



Strategy

Metacommunity:

$$\dot{N}_i = rN_i \left[1 - N_i - \sum_{j \neq i} A_{ij} N_j \right] + D[\langle N_i \rangle - N_i] + \sqrt{2T} N_i \eta_i(t),$$

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↓ $D \gg r$

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$T < D$

GLV

(low diversity)

$T > D$

θ -GLV

(high diversity iff $T > 2D$)

The $D \gg r$ adiabatic limit

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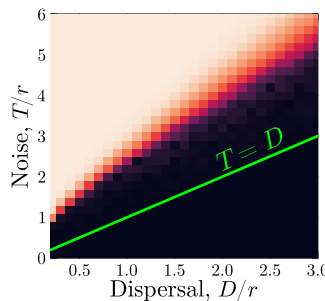
Valid whenever $D \gg r$

Valid only for $T < D$

Effective GLV dynamics for $T < D$

- Effective equation for means ($D - T \gg r$):

$$\partial_t \langle N_i \rangle = r \langle N_i \rangle \left[1 - \frac{D}{D - T} \langle N_i \rangle - \sum_{j \neq i} A_{ij} \langle N_j \rangle \right]$$



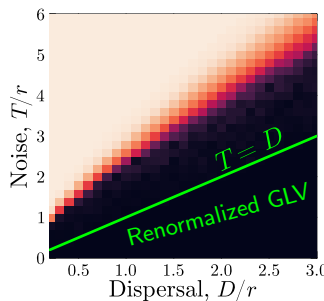
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$\Rightarrow$ *Standard GLV with renormalized carrying capacity*

$\Rightarrow S^* \in \mathcal{O}(1), \phi = 0 \checkmark$



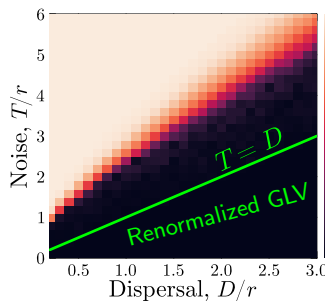
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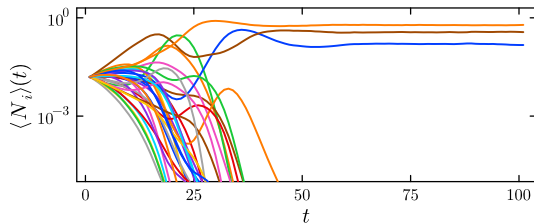
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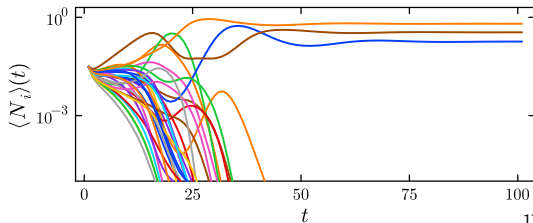
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Metacommunity ($D = 1, T = 0.1$):



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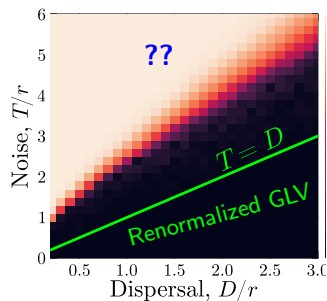
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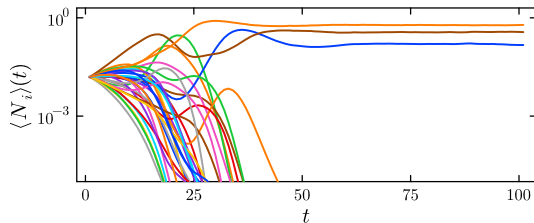
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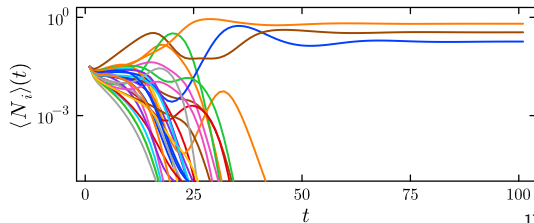
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Stationary abundance distribution in coexistence phase

$$\dot{N}_i = N_i [1 - N_i - \xi_i] + D[\langle N_i \rangle - N_i] + \sqrt{2T} N_i \eta_i(t), \quad \xi_i(t) \equiv \sum_{j \neq i} A_{ij} N_j(t)$$

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- **Coexistence phase:** Assume $\langle N_i \rangle > 0$, $\langle N_i \rangle \in \mathcal{O}(1/S)$
- **Claim:** for $D \gg r$, fluctuations of ξ_i vanish at large S :

$$\xi_i(t) \xrightarrow{S \rightarrow \infty} \langle \xi_i \rangle \equiv \sum_{j \neq i} A_{ij} \langle N_j \rangle$$

To show this:

1. Assume $\xi = \langle \xi \rangle$, find $\langle \xi \rangle$ self-consistently
2. Compute $\langle N_i^2 \rangle$
3. Show $\lim_{S \rightarrow \infty} \langle \xi^2 \rangle_c = 0$

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$$p_i(N) \propto e^{-\beta N} e^{-\beta D \langle N_i \rangle / N} N^{-\gamma_i}, \quad \gamma_i \equiv 2 + \beta(D + \langle \xi_i \rangle - 1).$$

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- Expand for small $\langle N_i \rangle \propto 1/S$ and solve for $\langle \xi \rangle$:

$$\boxed{\langle \xi_i \rangle = 1 + \mathcal{O}(S^{-\beta D})} \text{ for all } i.$$

Power law distribution of abundances

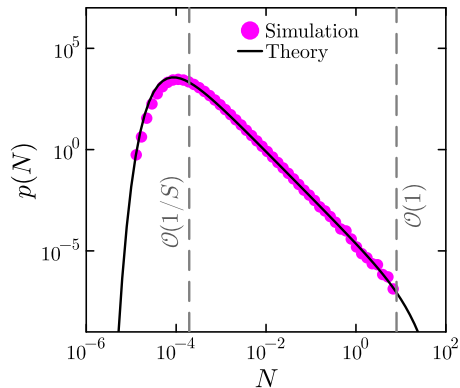
► Result:

All species follow the same power law

$$p_i(N) \propto e^{-\beta N} e^{-\beta D \langle N_i \rangle / N} N^{-2-\beta D},$$

► Power laws observed in natural ecosystems:

(Locey and Lennon, 2016; Ser-Giacomi, 2018; etc.)



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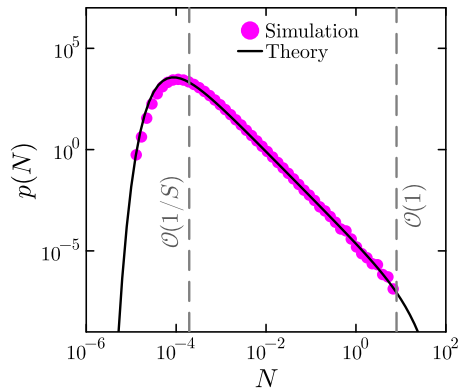
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► Consequence:

Anomalous scaling of moments



Anomalous scaling of moments

- For small $\langle N \rangle \propto 1/S$:

$$\langle N^2 \rangle \simeq \frac{D}{D-T} \langle N \rangle^2 \left[1 + (\dots) \langle N \rangle^{\beta D - 1} \right] + \mathcal{O}(\langle N \rangle^2)$$

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- Transition at $T = D$ (Swartz et al. 2022, Ottino-Löffler & Kardar 2020):

$$\langle N_i^2 \rangle \propto \begin{cases} \langle N_i \rangle^2, & T < D, \\ \langle N_i \rangle^{1+\beta D}, & T > D. \end{cases}$$

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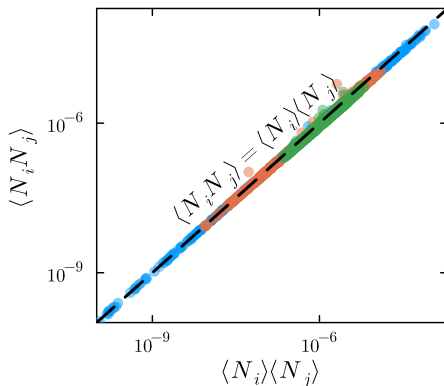
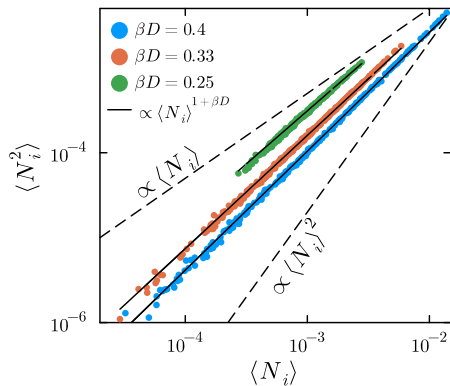
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- ▶ **Taylor's power law (1961):** $\langle N_i^2 \rangle_c \propto \langle N_i \rangle^{1+\theta}$

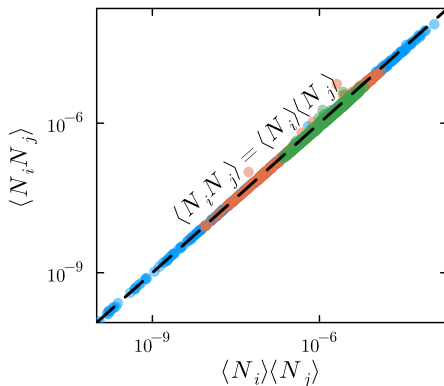
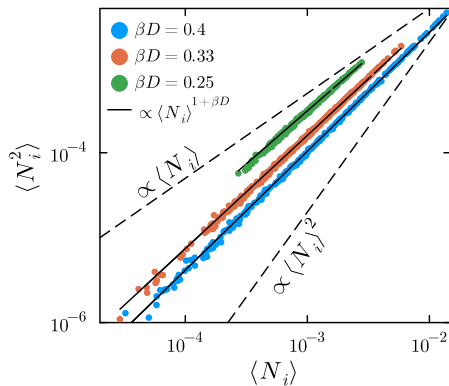
Widely verified in ecological data, nonuniversal exponent (Taylor 1961; Eisler et al. 2008)

Anomalous scaling of moments in the coexistence phase



Simulations support $\langle N_i^2 \rangle \propto \langle N_i \rangle^{1+\beta D}$ in coexistence phase.

Anomalous scaling of moments in the coexistence phase



Simulations support $\langle N_i^2 \rangle \propto \langle N_i \rangle^{1+\beta D}$ in coexistence phase.

$$\Rightarrow \left[\langle N_i \rangle - \langle N_i^2 \rangle - \sum_j A_{ij} \langle N_i N_j \rangle \right] \xrightarrow{D \gg r} \left[\langle N_i \rangle - \langle N_i \rangle^{1+\theta} - \sum_j A_{ij} \langle N_i \rangle \langle N_j \rangle \right]$$

θ -logistic Generalized Lotka-Volterra Equation

- Effective equation for means ($\theta \equiv \beta D$)

$$\langle \dot{N}_i \rangle = \langle N_i \rangle \left[1 - \langle N_i \rangle^\theta - \sum_{j \neq i} A_{ij} \langle N_j \rangle \right], \quad (\theta\text{-GLV}).$$

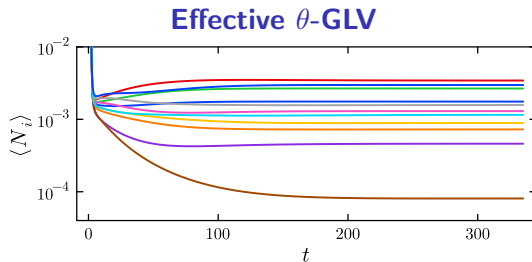
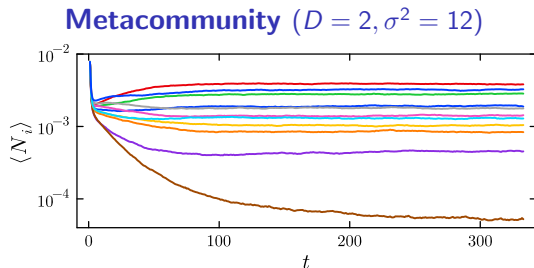
- Sometimes called **θ -logistic** self-regulation (Hatton et al.).

θ -logistic Generalized Lotka-Volterra Equation

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$$\langle \dot{N}_i \rangle = \langle N_i \rangle \left[1 - \langle N_i \rangle^\theta - \sum_{j \neq i} A_{ij} \langle N_j \rangle \right], \quad (\theta\text{-GLV}).$$

- Sometimes called **θ -logistic** self-regulation (Hatton et al.).

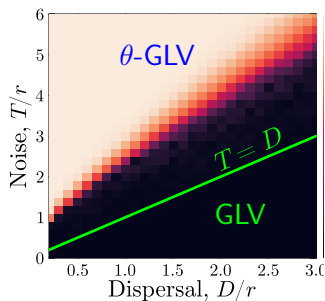


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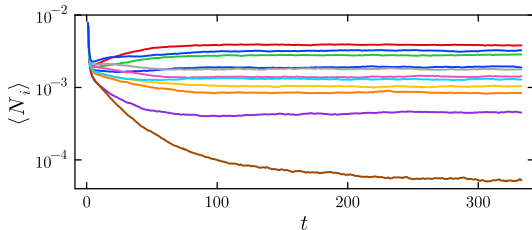
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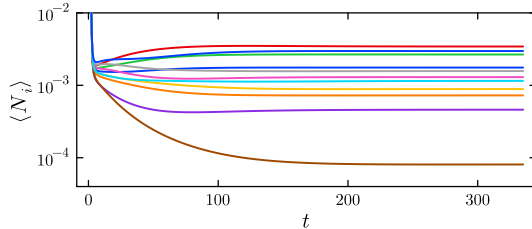
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Metacommunity ($D = 2, \sigma^2 = 12$)



Effective θ -GLV



Fixed points of the θ -GLV model

$$0 = \langle N_i \rangle \left[1 - \langle N_i \rangle^\theta - \sum_{j \neq i} A_{ij} \langle N_j \rangle \right], \quad \overline{A_{ij}} = \mu, \quad \overline{A_{ij} A_{kl}} = \sigma^2 \delta_{ik} \delta_{jl}.$$

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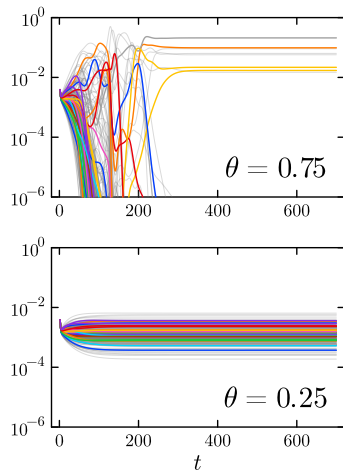
- ▶ Survival fraction $\phi = \mathbb{P}[\langle N \rangle > 0]$
- ▶ Solve using large- S ansatz: $\overline{\langle N \rangle} \sim S^{-\alpha}, \overline{\langle N \rangle}^2 \sim S^{-\gamma}, \phi \rightarrow 1$

Cavity solution of the θ -GLV model

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Results:

- There is a coexistence solution ($\phi = 1$) iff $\theta < \frac{1}{2}$

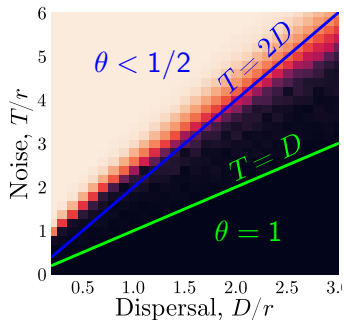


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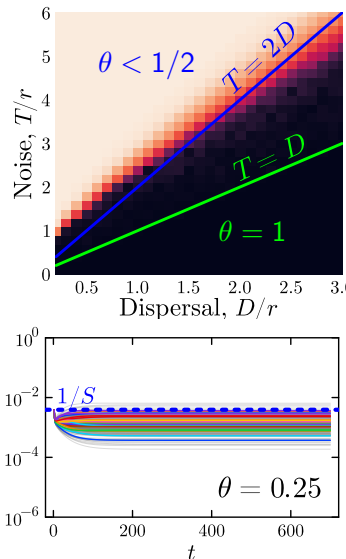
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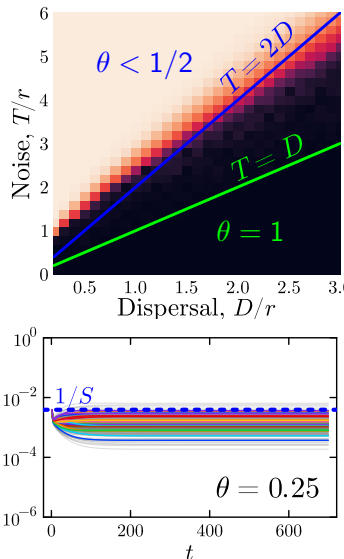
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- As $S \rightarrow \infty$, $\overline{\langle N \rangle^2} \rightarrow \overline{\langle N \rangle}^2$
 $\Rightarrow$ **disorder becomes irrelevant.**



What saves us from May's argument?

$$\langle \dot{N}_i \rangle = \langle N_i \rangle \left[1 - \langle N_i \rangle^\theta - \sum_{j \neq i} A_{ij} \langle N_j \rangle \right], \quad A_{ij} \sim \mathcal{N}(\mu, \sigma^2)$$

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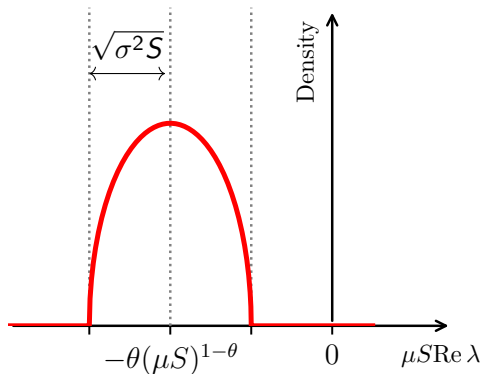
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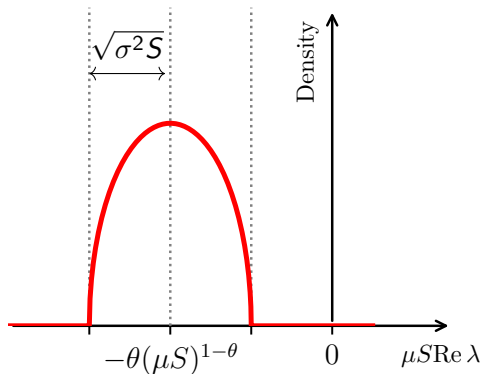
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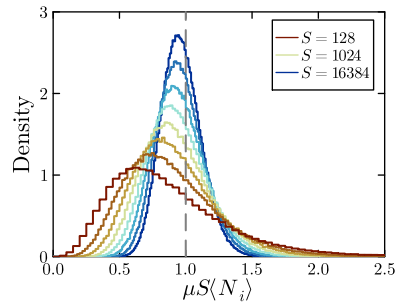


As $S \rightarrow \infty$: **All eigenvalues stable iff $\theta < 1/2$**

“Diversity begets stability” (Hatton, 2024)

Emergent neutrality

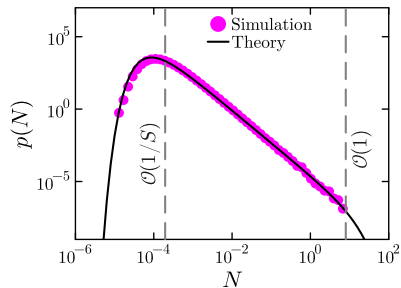
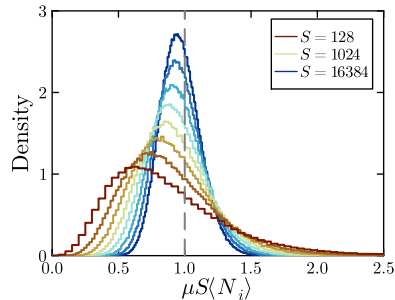
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$$p(N_1, \dots, N_S) \propto \prod_{i=1}^S e^{-\beta N_i} e^{-\beta D / \mu S N_i} N_i^{-2-\beta D}.$$

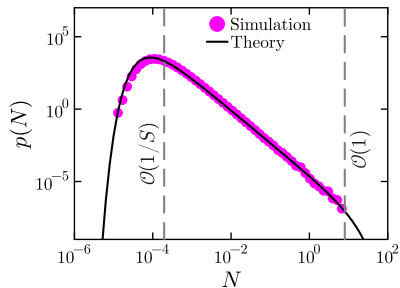
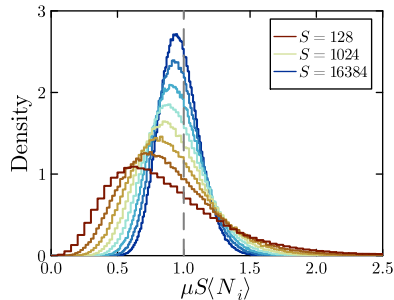


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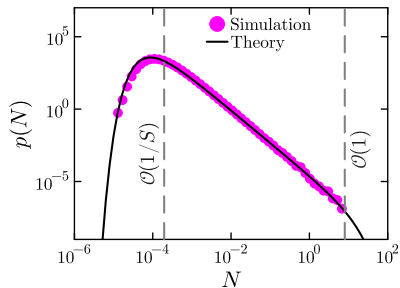
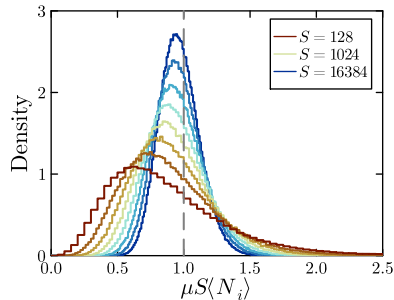
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Strongly-disordered pairwise interactions

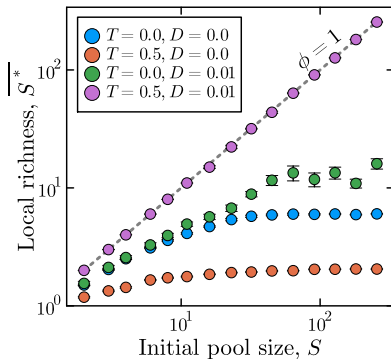
↓ $S \rightarrow \infty$

Community with indistinguishable species



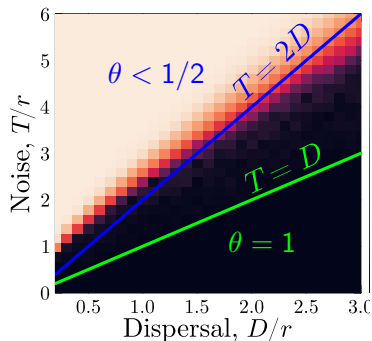
Conclusions

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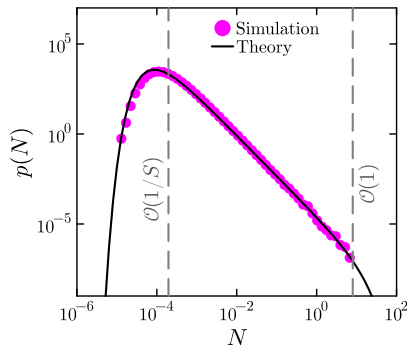
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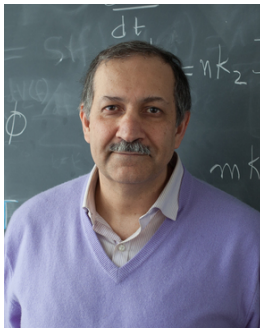


Acknowledgments

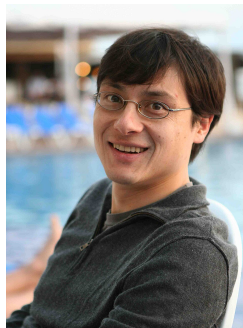
Dr. Daniel Swartz



Prof. Mehran Kardar



Prof. Jeff Gore



Funding: Tushar Shah and Sara Zion Physics Fellowship (MIT)

Nonequilibrium glass transition

$$\dot{N}_i = N_i \left[1 - N_i - \sum_j A_{ij} N_j \right] + D(\langle N_i \rangle - N_i), \quad A_{ij} \sim \mathcal{N}(\mu, \sigma^2)$$

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- **Environmental noise** $\propto N$ breaks FDT:

Allows nonequilibrium finite- T transition to novel “disordered” phase.

Fixed points of the θ -GLV model

- Rescale $M_i \equiv \langle N_i \rangle / K$, $A \rightarrow A/K$:

$$0 = M_i \left[1 - M_i^\theta - \sum_{j \neq i} A_{ij} M_j \right], \quad \overline{A_{ij}} = \mu, \quad \overline{A_{ij} A_{kl}} = \sigma^2 \delta_{ik} \delta_{jl}.$$

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$$\overline{M^n} = \frac{1}{\sqrt{2\pi\sigma^2\overline{S\overline{M}^2}}} \int_0^\infty dx x^{n/\theta} e^{-\frac{(x-1+\mu\overline{S\overline{M}})^2}{2\sigma^2\overline{S\overline{M}^2}}}.$$

Cavity solution of the θ -GLV model

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► Survival fraction: $\phi = \mathbb{P}[M > 0] = \frac{1}{\sqrt{2\pi}} \int_{-\frac{1-\mu S \bar{M}}{\sigma \sqrt{S \bar{M}^2}}}^{\infty} dx e^{-x^2/2}$

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► Survival fraction: $\phi = \mathbb{P}[M > 0] = \frac{1}{\sqrt{2\pi}} \int_{-\frac{1-\mu S \overline{M}}{\sigma^2 S \overline{M}^2}}^{\infty} dx e^{-x^2/2}$

► Large- S scaling ansatz:

$$\overline{M} \sim S^{-\alpha}, \quad \overline{M}^2 \sim S^{-\gamma}, \quad \frac{1 - \mu S \overline{M}}{\sigma^2 S \overline{M}^2} \simeq h S^{\eta}$$

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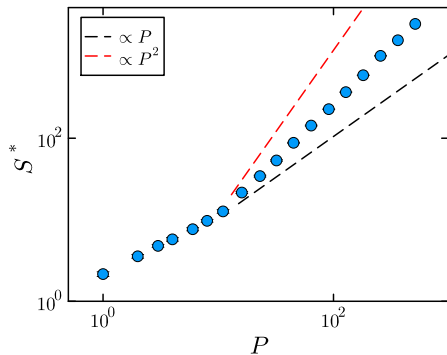
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► Determine α, γ, η by matching powers of S

Beyond fully-connected infinite networks

Finite P , $S \rightarrow \infty$ (fully-connected):



Grids and lines:

